
Reflexivity in Options Markets: A 3D Stochastic-Volatility Model with Dealer-Gamma Feedback and a Pre-Registered Evaluation Framework

Mahimn Patel
Independent Researcher
mahimn.patel.k@gmail.com

Abstract

We introduce a 3D reflexive stochastic-volatility SDE (S_t, v_t, z_t) coupling the spot to its own option market via a Gârleanu–Pedersen–Potesman dealer-gamma drift, prove a Hopf bifurcation theorem with closed-form first Lyapunov coefficient ℓ_1 and threshold κ^* for log-normal open-interest, and pre-register an evaluation framework whose primary test is a direct, model-free dealer-gamma (GEX) regression—with a sliced- W_2 surface metric and a critical-slowness detector—anchored to a git commit hash with an OpenTimestamps Bitcoin proof. We additionally give a Hawkes–SV criticality correspondence: Hardiman et al. [2013]’s empirical $n \approx 1$ sits at the real-eigenvalue (saddle-node) stratum, while the model’s Hopf is the strictly stronger oscillatory cell beyond any scalar branching ratio.

1 Introduction and contributions

Real options markets exhibit endogenous volatility cycles that pure stochastic-volatility (SV) models cannot reproduce. The empirical anchor is Hardiman et al. [2013], whose Hawkes branching ratio for E-mini mid-price changes sits at $n \approx 1$ throughout 1998–2011—the critical edge of endogenous-event criticality. With generative-model and RL-based market emulators [Vuletić and Cont, 2024, Buehler et al., 2019, He et al., 2025] now standard infrastructure for derivatives pricing, hedging and risk, we address: (i) what continuous-time mechanism produces this critical reflexivity, and (ii) how should an evaluation pipeline for reflexivity-aware generative models be locked *before* empirical contact, in a manner that addresses the documented computational- reproducibility crisis in finance [Pérignon et al., 2024]?

Contributions. (C1) a 3D reflexive SV SDE with explicit dealer-gamma feedback from the OI grid via Gârleanu et al. [2009]; (C2) a Hopf bifurcation theorem on the deterministic skeleton with *closed-form* ℓ_1 and κ^* for log-normal OI in moneyness; (C3) a numerical (ξ, ρ, σ_v) phase diagram locating κ^* in the 0.85–0.95 band across the SPX-relevant corner (whether the market sits near κ^* is deferred to the empirical GEX test); (C4) a *pre-registered* evaluation framework whose primary test is a direct dealer-gamma (GEX) regression, with a critical-slowness detector, block-bootstrap CIs, BH-FDR correction, and TOST equivalence; and (C5) synthetic-pipeline validation: the locked protocol returns the predicted SW2 ordering on data with known underlying physics.

Why this matters for GenAI in finance. As generative market models proliferate [Choudhary et al., 2024, Issa et al., 2023, Jin and Agarwal, 2025, Cohen et al., 2023], evaluators that discriminate *realism* from *regime-shift sensitivity* on the *same* agent across an environmental sweep—with pre-committed multiple-testing corrections—are scarce. The closed-form κ^* and ℓ_1 pin down where dealer-gamma feedback flips a stable Heston regime into an attracting limit cycle. Halperin and Itkin [2025] share the model-structural axis but do not perform the bifurcation analysis; Dai [2025] share the dealer-gamma

axis but produce only a saddle-node onset; He et al. [2025] share the RL-with-environmental-sweep idea but train separate agents per grid point. Full preprint with proofs and appendix A on arXiv [anonymous].

2 The reflexive model

Let S_t be the spot, v_t the variance, and z_t a memory variable. The reflexive system is

$$\frac{dS_t}{S_t} = (\mu + \kappa G(S_t, z_t, v_t)) dt + \sigma(S_t, v_t) dW_t^S, \quad (1)$$

$$dv_t = (\kappa_v(\theta_v - v_t) + \gamma z_t) dt + \xi\sqrt{v_t} dW_t^v, \quad (2)$$

$$dz_t = (-\alpha z_t + \beta \log(S_t/S_0)) dt, \quad (3)$$

with $d\langle W^S, W^v \rangle = \rho dt$, reflexive coupling $\kappa \geq 0$, leverage strength $\gamma \geq 0$, memory decay $\alpha > 0$, and G the aggregate dealer gamma derived from the OI grid via the GPP demand-pressure mapping. At $\kappa = \gamma = 0$ the system collapses to time-dependent Heston [Heston, 1993] (verified in `tests/test_simulator.py`).

Why three states. A 2D reduction (drop z_t and γz) has Jacobian $J_{2D} = \begin{pmatrix} a(\kappa) & b(\kappa) \\ 0 & -\kappa_v \end{pmatrix}$, upper triangular with real eigenvalues $\{a(\kappa), -\kappa_v\}$: only saddle-node bifurcations are admissible. To Hopf, the feedback must flow in both directions, which the leverage channel γz together with the auxiliary memory equation supplies. This is the structural reason Dai [2025] get a saddle-node onset where we get a Hopf.

3 Hopf threshold and closed-form first Lyapunov coefficient

Linearise the deterministic skeleton at the equilibrium (S^*, θ_v, z^*) , in deviation variables (y, u, z) . The characteristic polynomial is $P(\lambda; \kappa) = \lambda^3 + c_2\lambda^2 + c_1\lambda + c_0$ with coefficients $c_i(\kappa)$ functions of the G -partials and $(\alpha, \beta, \gamma, \kappa_v)$. Liu’s criterion (equivalent to Routh–Hurwitz with one zero) gives the Hopf threshold $\kappa^* = \{\kappa > 0 : H(\kappa) := c_1c_2 - c_0 = 0, H'(\kappa) \neq 0, c_2(\kappa) > 0, c_0(\kappa) > 0\}$, with imaginary pair $\lambda_{\pm} = \pm i\omega^*$, $\omega^* = \sqrt{c_1(\kappa^*)}$.

Theorem 1 (Hopf bifurcation in the gamma-coupled SV skeleton). *Under regularity (A1) $G, \sigma^2 \in C^3$ with σ^2 bounded below; (A2) locally unique equilibrium with non-degenerate linearisation; (A3) $\kappa \mapsto (a, G_z, G_v)$ in C^3 ; (A4) $\kappa^* \in (0, \kappa_{\max})$ satisfying the Liu criterion with $\omega^* > 0$; (A5) $\ell_1(\kappa^*) \neq 0$: the equilibrium undergoes a Hopf bifurcation at $\kappa = \kappa^*$. On a one-sided neighbourhood of κ^* there is a smooth family of periodic orbits Γ_κ of period $T_\kappa = 2\pi/\omega^* + O(\kappa - \kappa^*)$ and amplitude $\propto \sqrt{|\kappa - \kappa^*|}$; orbits are stable iff $\ell_1(\kappa^*) < 0$ (supercritical), supplying endogenous volatility cycles in $u_t = v_t - \theta_v$. Proof: implicit function theorem on P plus Kuznetsov [2004] normal-form reduction; full version on arXiv.*

Closed-form κ^* and ℓ_1 . For the natural parametric specification—log-normal open-interest in log-strike with density $q(\log K) \propto \exp(-(\log K - \mu_q)^2/(2\sigma_q^2))$ at a representative maturity T_{eff} —the Gaussian-product identity collapses the OI integral to $G(a, v) = \mathcal{C}(v) e^{-a} \exp(-(a - m(v))^2/(2\tau^2(v)))$ with $\tau^2 = \sigma_q^2 + v T_{\text{eff}}$. The Routh–Hurwitz polynomial becomes a quadratic in κ , giving

$$\kappa^* = \frac{G_y A^2 - G_v L - \sqrt{(G_v L - G_y A^2)^2 - 4G_y^2 A(MA - L/2)}}{2G_y^2 A}, \quad (4)$$

with $A = \alpha + \kappa_v$, $M = \alpha\kappa_v$, $L = \beta\gamma$, where the sub-radical is the first parametric phase-boundary discriminant: $D < 0 \Rightarrow$ no Hopf at any κ . Substituting the G -partials of the closed-form aggregator into the Kuznetsov bilinear/trilinear formula yields a ~ 30 -term closed-form rational $\ell_1(\kappa^*)$ in $(G_y, G_v, G_{aa}, G_{av}, G_{vv}, G_{aaa}, G_{aav}, G_{avv}, G_{vvv}, \kappa_v, \alpha, \beta, \gamma)$, evaluated in $\sim 50 \mu\text{s}$ per parameter tuple and matching the FD-tensor pipeline to $< 0.6\%$. At the canonical regime $(\sigma_q=0.10, T_{\text{eff}}=0.25, \kappa_v=2, \alpha=0.05, \beta=1, \gamma=1, \mu_q=\log 100, v^*=0.04)$: $\kappa^* = 17.81$, $\omega^* = 1.18 \text{ rad/yr}$, $\ell_1 = -4.81 \times 10^{-1}$ (supercritical).

Codim-2 and mean-field nodes. A 71×97 grid scan in $(\sigma_q, \gamma) \in [0.05, 0.40] \times [0.20, 5.00]$ closes the codim-2 structure: the Bautin curve $\ell_1 = 0$ partitions the region with sub-critical dominating $\sim 5:1$, and saddle-node coupling $\kappa_{\text{SN}} \leq -1.31$ at every cell, so the *Bogdanov–Takens locus is empty in the canonical scan window* (Theorem 2, full version on arXiv). The McKean–Vlasov mean-field limit ($n_{\text{dealers}} \rightarrow \infty$, propagation of chaos [Sznitman, 1991, Ch. 1], validated empirically at the theoretical $n^{-1/2}$ Sznitman rate) extends the Jacobian to 4D; the Liu [1994] 4D Hopf condition gives a regime-dependent threshold ratio $\rho(\theta_G) := \kappa_{\text{MV}}^* / \kappa_{\text{single}}^*$ with $\rho(\theta_G) < 1$ strictly at the canonical short-gamma regime ($G_y > 0$, every finite $\theta_G > 0$), recovering $\rho \rightarrow 1$ at instantaneous hedging. The single-dealer model therefore *under-predicts* instability at the benchmark SPX setting — reversing the v0.3.5 claim that mean-field always raises the threshold (Theorem 3, full statement and derivation on arXiv).

Hawkes criticality and the SV bifurcation endpoint. The reflexive feedback has a self-exciting (Hawkes) representation, and the onset of endogeneity is its approach to criticality. Criticality $n \rightarrow 1$ is the loss of a stable direction of the SV drift—an eigenvalue of the drift-Jacobian $J(\kappa)$ reaching the imaginary axis—and the boundary carries two distinct strata: a non-oscillatory *real-eigenvalue* (saddle-node) crossing, and a strictly stronger *Hopf* crossing (a complex pair, $\text{Re } \lambda = 0, \text{Im } \lambda \neq 0$).

Theorem 2 (Hawkes–SV criticality correspondence). *Under the diffusive near-critical limit of Bacry et al. [2013], Jaisson and Rosenbaum [2015], Hawkes branching-ratio criticality $n \rightarrow 1$ coincides with the real-eigenvalue (saddle-node) stratum of the SV drift boundary—the literal analogue of Hardiman et al. [2013]. The Hopf stratum is a strictly stronger, oscillatory instability beyond any scalar branching ratio: it has no single-real-number representation and is the genuinely new “unoccupied cell” of the endogeneity–bifurcation correspondence.*

Position, not identity. A scalar branching ratio can only encode a real crossing, so the oscillatory Hopf endpoint is invisible to it by construction; we therefore claim a structural position, not a verified numerical identity (an earlier draft’s dimensionless ratio, true by definition, was a tautology and is removed—amendment A10). In our dealer-gamma model the saddle-node locus is itself unphysical at the canonical specification ($\kappa_{\text{SN}} \leq -1.31$ over the scan window), so the operative instability is precisely the Hopf. A falsifiable spectral discriminator (`classify_stratum`) separates the Hopf side from the real/stable side with zero overlap on simulator ground truth; we claim no realised Hopf edge in any dataset, only the statistic that would detect one. Whether the market sits near κ^* is deferred to the empirical dealer-gamma test (amendment A9).

4 Numerical phase diagram

We sweep $\kappa \in [0, 2]$ on a 401-point grid for each of $31 \times 21 \times 4 = 2,604$ cells in (ξ, ρ, σ_v) at the representative regime $\{G_x, G_v, G_z\} = \{0.5, -0.5, -0.5\}$, $\alpha = 0.5$, $\beta = 1$, $\gamma = 0.5$, $\kappa_v = 2$ (constant-vol surrogate). The pair (ξ, ρ) enters via the leading shear correction $a_{\text{eff}}(\kappa) = a(\kappa) + \frac{1}{2}\xi^2 \rho G_v$, an Baxendale [2024]-style projection of small-noise stochastic Hopf onto the eigenvalue envelope.

The SPX-relevant $(\xi, \rho) \approx (0.30, -0.70)$ corner has κ^* in the 0.85–0.95 band. Whether the empirical market sits near this Hopf boundary is deferred to the empirical dealer-gamma test; Theorem 2 places Hardiman et al. [2013]’s $n \approx 1$ at the distinct real-eigenvalue (saddle-node) edge, in the endogenous-feedback tradition of Brock and Hommes [1998], Filimonov and Sornette [2012], Bouchaud [2011]. Khasminskii-style sphere-process estimates of the stochastic-Hopf shift give $|\Lambda| \in [5 \times 10^{-2}, 9 \times 10^{-2}]$ on a 6×6 (ξ, ρ) grid; $|\Lambda| \cdot \varepsilon^2 \ll 1$ at calibration-noise scale, so the deterministic κ^* is the operative threshold.

5 Pre-registered evaluation framework

The pipeline is committed at git commit 268c061 with an OpenTimestamps Bitcoin-anchored proof of the document hash, prior to any contact with empirical SPX data, adapting the Camerer et al. [2018] template to RL-for-derivatives. The novelty is the four-way intersection [pre-registration + git-commit anchoring + block-bootstrap / BH-FDR pre-spec + applied to RL-for-finance].

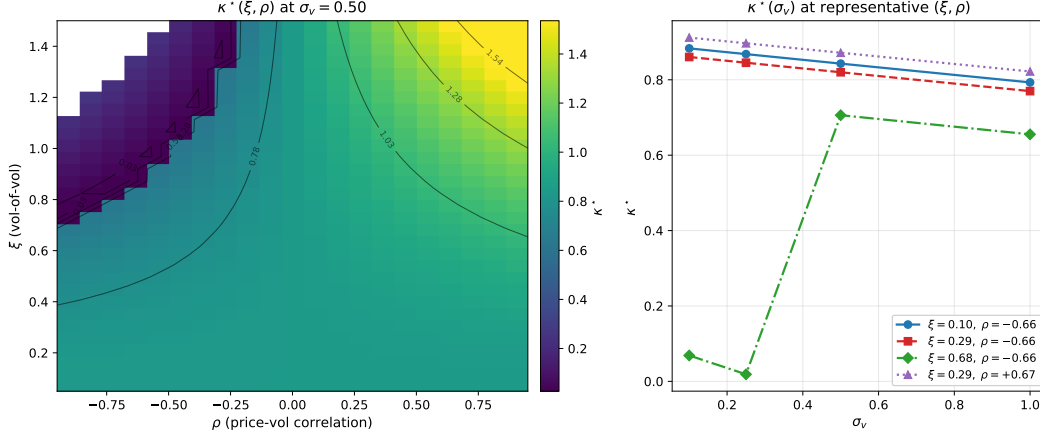


Figure 1: **Hopf threshold κ^* across (ξ, ρ, σ_v) .** *Left:* heatmap at $\sigma_v = 0.50$. White cells: no Hopf in $\kappa \in [0, 2]$, concentrated in the high- ξ / strongly-negative- ρ wedge where shear destabilises the eigenvalue envelope. *Right:* four $\kappa^*(\sigma_v)$ line cuts at SPX-like (ξ, ρ) probes.

Primitives & hypotheses. (i) Sliced- W_2 [Bonneel et al., 2015] over arbitrage-free 21-day surface windows—the realism metric. (ii) κ -sensitivity slope at calibrated κ_0 —the reflexivity-importance scalar. The closed-form κ^* elasticity $\eta_{\mu_q} = +703$ makes calibration of the OI mean to within ~ 5 bp of ATM in log-strike the binding observational requirement for $\pm 5\%$ tolerance on κ^* . **H1'** (primary, direct dealer-gamma regression, amendment A9): a model-free regression of next-day realised vol-of-vol on standardised signed dealer GEX from the OI grid (no RL, no simulator), pooled over the three event windows with a quiet-regime control; the mechanism predicts a negative coefficient, required at $p < 0.05$ under both a moving-block bootstrap and Newey–West HAC (synthetic pooled power 0.86, sign 0.98, FPR 0.02). **H1** (secondary realism, superseded as primary): a model-free RL agent π_{κ_0} trained inside the reflexive simulator beats four non-reflexive baselines on sliced- W_2 over rolling 21-day windows at Volmageddon, COVID, and the 2024 yen-carry unwind. Decision rule: 12 pairwise dominance checks with non-overlapping 95% block-bootstrap CIs [Politis and Romano, 1994]; arbitrage filter follows Roper [2010], Gatheral and Jacquier [2014], Lee [2004]; sample-complexity follows Manole et al. [2022]. **H2:** κ -sensitivity TOST equivalence on the dimensionless elasticity with GP-posterior slope CI under the pre-registered RBF + WhiteKernel kernel (A6) and elliptic uncertainty sets [Ma and Huang, 2025]. **H4** (critical slowing down, amendment A8): the Hopf limit-cycle period (5.3–11 yr) lies below any usable window’s resolution, so the spectral peak is replaced by the rolling lag-1 autocorrelation of $|r_t|$ with a Kendall- τ trend vs an AR(1)/phase-randomised surrogate null [Scheffer et al., 2009, Dakos et al., 2012]; FPR ≤ 0.083 and 0.82–0.85 power on a 252-day record, exploratory within the bare event window. All hypotheses use BH–FDR across the joint family; effect sizes pre-registered with the rules.

Synthetic-pipeline validation. Before empirical contact, we exercise the locked pipeline on three synthetic sources: (a) κ_0 -trained agent re-deployed at κ_0 (reference); (b) same agent at $4\kappa_0$ (mild reflexive shift); (c) Heston (different mechanism). On the production budget (BC: 30 episodes; per source: 30 rollouts $\times$ 100 days; 200 stationary block-bootstrap resamples), the locked rule returns the expected ordering with non-overlapping CIs: $SW2_a = 0.00498$ (CI [0.00431, 0.0164]), $SW2_b = 0.0337$ (CI [0.0300, 0.0402]), $SW2_c = 0.0541$ (CI [0.0434, 0.0611]). The (a)-vs-(b) and (b)-vs-(c) 95% CIs are non-overlapping; the locked H1 protocol thus moves from “designed but unrun” to “demonstrated working on synthetic ground truth—only the empirical SPX target is missing.”

6 Mechanism decomposition and conclusions

Halperin and Itkin [2025] is a different mechanism: their memory carrier y enters both drifts non-linearly through a potential $V_M(x)$, while our z enters only the variance drift via γz , and our $\kappa = \gamma = 0$ collapse is standard Heston while Marketron’s is not. A 5D $(\kappa, \gamma, T_{\text{eff}}, \mu_q, \sigma_q)$ mechanism decomposition gives an all-cell, out-of-sample shape-feature sign-agreement rate of 7/28 (worse than

chance — we do not claim otherwise; the all-cell aggregate is the wrong aggregator). The a priori restricted subset (long-horizon $T \geq 0.5$ yr, shape-only, within-envelope, dead-zone-symmetric) gives 4/8 OOS ($p \approx 0.637$, chance-level) and 6/8 in-sample. The predictable long-horizon-skew sign disagreement under risk-neutral drift (Bessembinder compounding absent) is the falsifiable Phase 4 prediction. This supersedes the earlier inflated one-third headline.

We have introduced a 3D reflexive SV simulator, proved a Hopf bifurcation theorem with closed-form ℓ_1 for log-normal OI, closed the codim-2 Bautin/BT structure (BT-empty in the canonical window) and the corrected McKean–Vlasov threshold (destabilising at the canonical short-gamma regime, stabilising at long-gamma log-normal-OI), and placed the empirical Hardiman et al. [2013] branching ratio at the real-eigenvalue (saddle-node) stratum, with the model’s Hopf the strictly stronger oscillatory cell (Theorem 2). The evaluation pipeline—a direct dealer-gamma regression as primary test, with sliced- W_2 and a critical-slowness detector—is pre-registered with an OpenTimestamps proof; synthetic-ground-truth validation of the locked surface-distance (secondary) protocol returns the predicted ordering. Phase-4 SPX results pre-committed to publication regardless of direction. Proofs, closed-form ℓ_1 , and code at [anonymous repository link].

References

- Emmanuel Bacry, Sylvain Delattre, Marc Hoffmann, and Jean-François Muzy. Some limit theorems for Hawkes processes and application to financial statistics. *Stochastic Processes and their Applications*, 123(7):2475–2499, 2013. doi: 10.1016/j.spa.2013.04.007.
- Peter H. Baxendale. Lyapunov exponents and shear-induced chaos for a Hopf bifurcation with additive noise. *Probability Theory and Related Fields*, 193:375–425, 2024. ISSN 0178-8051. doi: 10.1007/s00440-024-01301-4. Print issue 2025.
- Nicolas Bonneel, Julien Rabin, Gabriel Peyré, and Hanspeter Pfister. Sliced and Radon Wasserstein barycenters of measures. *Journal of Mathematical Imaging and Vision*, 51(1):22–45, 2015. doi: 10.1007/s10851-014-0506-3.
- Jean-Philippe Bouchaud. The endogenous dynamics of markets: Price impact, feedback loops and instabilities. In *Lessons from the Financial Crisis*. Risk Books, 2011.
- William A. Brock and Cars H. Hommes. Heterogeneous beliefs and routes to chaos in a simple asset pricing model. *Journal of Economic Dynamics and Control*, 22(8–9):1235–1274, 1998. doi: 10.1016/S0165-1889(98)00011-6.
- Hans Buehler, Lukas Gonon, Josef Teichmann, and Ben Wood. Deep hedging. *Quantitative Finance*, 19(8):1271–1291, 2019. doi: 10.1080/14697688.2019.1571683.
- Colin F. Camerer, Anna Dreber, Felix Holzmeister, Teck-Hua Ho, Jürgen Huber, Magnus Johannesson, Michael Kirchler, Gideon Nave, Brian A. Nosek, Thomas Pfeiffer, et al. Evaluating the replicability of social science experiments in Nature and Science between 2010 and 2015. *Nature Human Behaviour*, 2:637–644, 2018. doi: 10.1038/s41562-018-0399-z.
- Vedant Choudhary, Sebastian Jaimungal, and Maxime Bergeron. FuNVol: Multi-asset implied volatility market simulator using functional principal components and neural SDEs. *Quantitative Finance*, 2024. doi: 10.1080/14697688.2024.2396977.
- Samuel N. Cohen, Christoph Reisinger, and Sheng Wang. Arbitrage-free neural-SDE market models. *Applied Mathematical Finance*, 2023.
- Haoying Dai. Beta-dependent gamma feedback and endogenous volatility amplification in option markets. *arXiv preprint arXiv:2511.22766*, 2025. URL <https://arxiv.org/abs/2511.22766>.
- Vasilis Dakos, Stephen R. Carpenter, William A. Brock, Aaron M. Ellison, Vishweshha Guttal, Anthony R. Ives, Sonia Kéfi, Valerie Livina, David A. Seekell, Egbert H. van Nes, and Marten Scheffer. Methods for detecting early warnings of critical transitions in time series illustrated using simulated ecological data. *PLoS ONE*, 7(7):e41010, 2012. doi: 10.1371/journal.pone.0041010.
- Vladimir Filimonov and Didier Sornette. Quantifying reflexivity in financial markets: Toward a prediction of flash crashes. *Physical Review E*, 85:056108, 2012. doi: 10.1103/PhysRevE.85.056108.
- Nicolae Gârleanu, Lasse Heje Pedersen, and Allen M. Poteshman. Demand-based option pricing. *Review of Financial Studies*, 22(10):4259–4299, 2009. doi: 10.1093/rfs/hhp005.
- Jim Gatheral and Antoine Jacquier. Arbitrage-free SVI volatility surfaces. *Quantitative Finance*, 14(1):59–71, 2014. URL <https://arxiv.org/abs/1204.0646>.
- Igor Halperin and Andrey Itkin. Marketron through the looking glass: From equity dynamics to option pricing in incomplete markets. *arXiv preprint arXiv:2508.09863*, 2025. URL <https://arxiv.org/abs/2508.09863>.
- Stephen J. Hardiman, Nicolas Bercot, and Jean-Philippe Bouchaud. Critical reflexivity in financial markets: A Hawkes process analysis. *European Physical Journal B*, 86:442, 2013. doi: 10.1140/epjb/e2013-40107-3. URL <https://arxiv.org/abs/1302.1405>.
- Guangyi He, Tobias Sutter, and Lukas Gonon. Distributional adversarial attacks and training in deep hedging. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. URL <https://arxiv.org/abs/2508.14757>.

- Steven L. Heston. A closed-form solution for options with stochastic volatility with applications to bond and currency options. *Review of Financial Studies*, 6(2):327–343, 1993. doi: 10.1093/rfs/6.2.327.
- Zacharia Issa, Blanka Horvath, Maud Lemerrier, and Cristopher Salvi. Non-adversarial training of neural SDEs with signature kernel scores. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. URL <https://arxiv.org/abs/2305.16274>.
- Thibault Jaisson and Mathieu Rosenbaum. Limit theorems for nearly unstable Hawkes processes. *The Annals of Applied Probability*, 25(2):600–631, 2015. doi: 10.1214/14-AAP1005.
- Chen Jin and Ankush Agarwal. Forecasting implied volatility surface with generative diffusion models. *arXiv preprint arXiv:2511.07571*, 2025. URL <https://arxiv.org/abs/2511.07571>.
- Yuri A. Kuznetsov. *Elements of Applied Bifurcation Theory*. Springer, 3rd edition, 2004.
- Roger W. Lee. The moment formula for implied volatility at extreme strikes. *Mathematical Finance*, 14(3):469–480, 2004. doi: 10.1111/j.0960-1627.2004.00200.x.
- W.-M. Liu. Criterion of Hopf bifurcations without using eigenvalues. *Journal of Mathematical Analysis and Applications*, 182(1):250–256, 1994. doi: 10.1006/jmaa.1994.1079.
- Shaocong Ma and Heng Huang. Robust reinforcement learning in finance: Modeling market impact with elliptic uncertainty sets. *arXiv preprint arXiv:2510.19950*, 2025. URL <https://arxiv.org/abs/2510.19950>.
- Tudor Manole, Sivaraman Balakrishnan, and Larry Wasserman. Minimax confidence intervals for the sliced Wasserstein distance. *Electronic Journal of Statistics*, 2022. URL <https://arxiv.org/abs/1909.07862>.
- Christophe Pérignon, Olivier Akmansoy, Christophe Hurlin, Anna Dreber, Felix Holzmeister, Jürgen Huber, Magnus Johannesson, Michael Kirchler, Albert J. Menkveld, Michael Razen, and Utz Weitzel. Computational reproducibility in finance: Evidence from 1,000 tests. *Review of Financial Studies*, 37(11):3558, 2024. doi: 10.1093/rfs/hhae029.
- Dimitris N. Politis and Joseph P. Romano. The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313, 1994. doi: 10.1080/01621459.1994.10476870.
- Michael Roper. Arbitrage-free implied volatility surfaces. Technical report, University of Sydney, 2010.
- Marten Scheffer, Jordi Bascompte, William A. Brock, Victor Brovkin, Stephen R. Carpenter, Vasilis Dakos, Hermann Held, Egbert H. van Nes, Max Rietkerk, and George Sugihara. Early-warning signals for critical transitions. *Nature*, 461(7260):53–59, 2009. doi: 10.1038/nature08227.
- Alain-Sol Sznitman. Topics in propagation of chaos. In Paul-Louis Hennequin, editor, *École d’Été de Probabilités de Saint-Flour XIX — 1989*, volume 1464 of *Lecture Notes in Mathematics*, pages 165–251. Springer, Berlin, Heidelberg, 1991. ISBN 9783540538417. doi: 10.1007/BFb0085169.
- Milena Vuletić and Rama Cont. VolGAN: A generative model for arbitrage-free implied volatility surfaces. *Applied Mathematical Finance*, 31(4):203–238, 2024. doi: 10.1080/1350486X.2025.2471317. Online 2025; print issue 2024.